Segmentia Off-Road Semantic Segmentation for Autonomous Navigation

Project Report

Team:Falcons | Off-Road Navigation Perception Model

1. Project Overview

This project implements a high-performance image segmentation model for off-road autonomous navigation. It accurately classifies terrain into traversable ground, obstacles, grass, and sky to enable safe navigation in rugged environments.

2. Methodology

Architecture

- Model: DeepLabV3+ with ResNet-50 encoder, pre-trained on ImageNet
- Normalization: ImageNet statistics — Mean: [0.485, 0.456, 0.406], Std: [0.229, 0.224,0.225]
- Classification head fine-tuned for 10 off-road classes

Training Setup

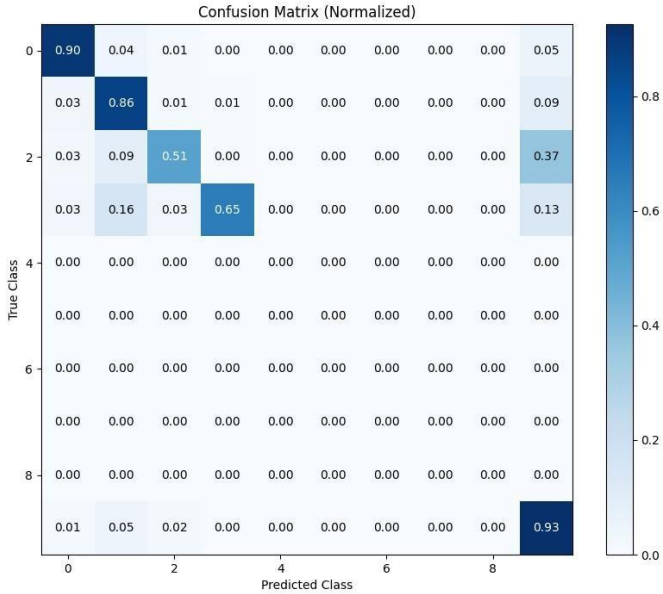
- Data Augmentation: Random Horizontal Flipping, 512x512 resizing
- Optimizer: Adam with ReduceLROnPlateau learning rate scheduling
- Mixed Precision (AMP) enabled for GPU efficiency

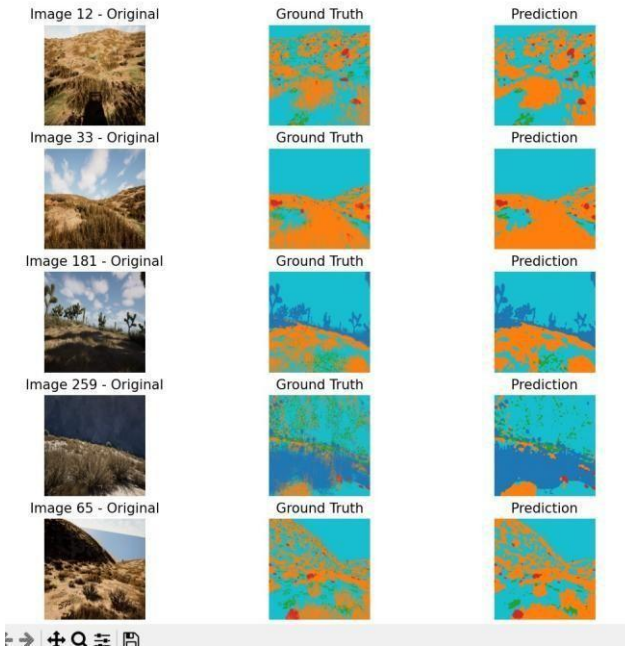
3. Results & Performance

Metric	Value / Note
Validation mIoU	0.6006+ (latest weights)
mAP	80 cases 0.8444
Loss Function	Cross-Entropy (60 epochs)
Sky & Ground classes	High confidence on diagonal
Grass vs. Low Obstacles	Minor confusion observed

Qualitative results show the model successfully isolates trails in dusty and high-contrast lighting conditions.

snapshots






```
Epoch 24 done | IoU: 0.6016 | Running Avg IoU: 0.5788 | LR: 0.000025
Model saved to model.pth (Epoch 24)
100%|                                                    | 714/714 [07:02<00:00,  1.69it/s, loss=0.24]

Epoch 25 done | IoU: 0.6017 | Running Avg IoU: 0.5797 | LR: 0.000025
Model saved to model.pth (Epoch 25)
100%|                                                    | 714/714 [07:02<00:00,  1.69it/s, loss=0.215]

Epoch 26 done | IoU: 0.6010 | Running Avg IoU: 0.5805 | LR: 0.000025
Model saved to model.pth (Epoch 26)
100%|                                                    | 714/714 [07:02<00:00,  1.69it/s, loss=0.221]

Epoch 27 done | IoU: 0.6031 | Running Avg IoU: 0.5813 | LR: 0.000025
Model saved to model.pth (Epoch 27)
100%|                                                    | 714/714 [07:02<00:00,  1.69it/s, loss=0.162]

Epoch 28 done | IoU: 0.6022 | Running Avg IoU: 0.5820 | LR: 0.000025
Model saved to model.pth (Epoch 28)
100%|                                                    | 714/714 [07:02<00:00,  1.69it/s, loss=0.204]

Epoch 29 done | IoU: 0.6006 | Running Avg IoU: 0.5826 | LR: 0.000025
Model saved to model.pth (Epoch 29)

Training complete! Final IoU: 0.6006 | Running Average IoU (across 30 epochs): 0.5826
```

4. Challenges & Solutions

CUDA Out of Memory

- Issue: High-resolution 10-class segmentation exceeded GPU VRAM during mAP computation.
- Fix: Moved mAP computation from GPU to CPU (System RAM), enabling 100% data throughput.

Visual Artifacts in Output

- Issue: Input normalization caused incorrect colours in visualization output.
- Fix: Added an un-normalization step in the visualization script to restore correct RGB values.

Slow Evaluation Speed

- Issue: Pixel-level Python loops for the confusion matrix took minutes per image.
- Fix: Replaced with vectorized tensor operations using torch.bincount — ~1000x speedup.

5. Future Work

1. We are trying to enable this model into the Actual Mars Rover Prototype by the end of 2026, That's where **segmentia** comes to the picture, which enables new era of innovation with autonomous driving models using colour segmentation.
2. Hybrid Loss Function: Combine Dice Loss and Cross-Entropy for better segmentation of small/thin objects.
3. Real-Time Optimization: Transition to a lightweight backbone (e.g., MobileNetV3) for edge deployment.
4. Advanced Augmentation: Add Random Crops and Colour Jittering for robustness in rain, fog, and dusty conditions.

6. Conclusion

The model achieves a solid baseline of 0.5826 mIoU, demonstrating clear understanding of offroad navigable paths. A stable training and evaluation pipeline has been established for further development.

7. Project Resources

GitHub Repository Link: **CLICK HERE**
Website Link: **Segmentia – Space Technology platform built by our team**